

Dynamic Pricing to Minimize Maximum Regret

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We consider a dynamic pricing problem that involves selling a given inventory of a single product over a short, two-period selling season. There is insufficient time to replenish inventory during this season, hence sales are made entirely from inventory. The demand for the product is a stochastic, nonincreasing function of price. We assume interval uncertainty for demand, that is, knowledge of upper and lower bounds but not a probability distribution, with no correlation between the two periods. We minimize the maximum total regret over the two periods that results from the pricing decisions. We consider a dynamic model where the decision maker chooses the price for each period contingent on the remaining inventory at the beginning of the period, and a static model where the decision maker chooses the prices for both periods at the beginning of the first period. Both models can be solved by a polynomial time algorithm that solves systems of linear inequalities. Our computational study demonstrates that the prices generated by both our models are insensitive to errors in estimating the demand intervals. Our dynamic model outperforms our static model and two classical approaches that do not use demand probability distributions, when evaluated by maximum regret, average relative regret, variability, and risk measures. Further, our dynamic model generates a total expected revenue which closely approximates that of a maximum expected revenue approach which requires demand probability distributions.

Key words: dynamic pricing; interval uncertainty; minimax regret; linear programming

History: Received: July 2014; Accepted: July 2016 by William Cooper, after 4 revisions.

1. Introduction

Numerous successful real-world applications of dynamic pricing are motivating growing academic research interest. We consider a dynamic pricing problem where the decision maker (DM) is a retailer that sells a given inventory of a single product over a short selling season consisting of two time periods, a regular sales period and a clearance period, without inventory replenishment.

Talluri and van Ryzin (2004) review the theory and practice of revenue management. A recent survey article (Chen and Chen 2015) identifies many dynamic pricing problems in revenue management applications. These applications arise in, for example, fashion apparel (Caro and Gallien 2012) and consumer electronics (Kuo et al. 2011). These industries are characterized by a selling season that is short (e.g., 4–8 weeks). This does not allow sufficient time for inventory replenishment. Also, it results in limited sales data availability, and hence unavailability of a complete probability distribution for demand at a given price. A more realistic assumption about

demand information is *interval uncertainty* (Shary 2002), that is, a lower and upper bound on demand at a given price. This modeling assumption can easily be justified from management practice.

Practitioners often characterize demand using a small number of scenarios, for example, low and high demand scenarios, which naturally leads to an interval representation of uncertain demand. For a product without sales history, practitioners often use sales data from the same product family to estimate possible demand scenarios (Gilliland and Guseman 2009). Many researchers used intervals to model demand uncertainty; examples include Kouvelis and Yu (1997) for robust discrete optimization problems, Lan et al. (2008) and Perakis and Roels (2010) for robust revenue management problems, Thiele (2009) for a robust static pricing problem, and Li et al. (2011) for robust network planning. There are also works (e.g., Adida and Perakis 2010, Perakis and Sood 2006) that study dynamic pricing problems, where demand uncertainty is modeled by intervals or more general uncertainty sets. However, in these dynamic pricing problems, the functional form of demand is assumed

to be known. Whereas, in the problems we study, even the functional form is unknown.

We apply the widely used criterion under a high level of uncertainty of *minimizing the maximum regret*, that is, *minimax regret*, developed by Savage (1951). Regret is the difference between the total revenue of an ideal (i.e., clairvoyant) pricing solution and that of the actual solution used by the DM. The following discussion justifies the use of this criterion. First, Schlag (2006) provides a persuasive foundational case for the minimax regret criterion. He identifies minimax regret as the unique criterion that satisfies several important axioms of decision making in the absence of prior probabilities. Second, empirical evidence supports this criterion as a predictor of decision making behavior, for example in insurance markets (Razin 1976). This criterion is less conservative than another commonly used criterion, maximizing the worst-case total revenue, that is, *maximin*. Third, many firms set explicit or implicit revenue targets based on an ideal situation. Using the minimax regret criterion guarantees that the resulting solution is not too far from the ideal solution. Fourth, the minimax regret criterion is used extensively, see, for example, Perakis and Roels (2008) on the newsvendor problem with partial information, Perakis and Roels (2010) on network revenue management under interval uncertainty, and Besbes and Zeevi (2009) and Wang et al. (2014) on dynamic pricing problems with demand learning.

As discussed by Chen and Chen (2015), most of the literature on dynamic pricing assumes that (i) demand as a function of price is either deterministic, or stochastic with a known distribution, and (ii) the DM is risk neutral with a goal of maximizing the total expected profit. Recently however, dynamic pricing problems where the demand distribution is not fully known have received more attention (Chen and Chen 2015). Such problems can be classified into *parametric problems* where the functional form of demand is known to the decision maker, but some underlying parameters are unknown; and *nonparametric problems* where even the functional form of demand is unknown. In the latter case, the DM only knows an uncertainty set for all demand realizations, or some regularity conditions for demand. For many products such as fashion apparel with a short selling season, it is not possible for the DM to identify even the functional form of the demand. Thus, a nonparametric model is more realistic.

We consider two models of the problem of minimizing the maximum regret under interval uncertainty, and design efficient algorithms to find optimal solutions for both models. In the first *dynamic* model, the DM (“he”) chooses the price for each period

contingent on the inventory at the beginning of the period. More specifically, at the beginning of the first period, he chooses a price for the product. Given that price, a clairvoyant adversary (AD, “she”), fixes the demand for every possible price, including for the price chosen by the DM, subject to some constraints. Consequently, the DM knows his remaining inventory after the first period. The same sequence of decisions and events repeats in the second period. The AD solves the same problem in each period with the knowledge of the DM’s price in this period, with the objective of finding a solution that maximizes the DM’s total regret over the two periods. The DM finds a solution that minimizes the maximum value of his total regret. In the second *static* model, the DM chooses the prices for both periods at the beginning of the first period without considering possible demand in the first period. After observing the DM’s prices for both periods, the AD makes all her decisions together in the first period with the same objective as in the first model. In the dynamic model the DM has more information in making his pricing decision for the second period, whereas the AD has less information in making her decisions, than in the static model. Therefore, the DM has a less maximum total regret under the dynamic model than under the static model.

Our computational study generates several managerial insights. The optimal prices generated by our models are insensitive to the widths of the demand intervals. Hence, even if the upper and lower bounds of the demand intervals are poorly estimated, our models still generate optimal or near-optimal prices. Our dynamic model outperforms our static model and three other commonly used approaches, as measured by the maximum and the average relative regret value. Further, our dynamic model generates a total expected revenue that is very close to that of a maximum expected revenue approach. One benchmark approach, which, like our models, does not require knowledge of the demand probability distributions, outperforms our models in terms of the total expected revenue measure. However, our dynamic model outperforms this benchmark approach on commonly used variability and risk measures. Thus, our dynamic model requires little demand information but provides robust solutions.

The remainder of this study is organized as follows. In section 2, we review the related literature. In section 3, we describe our notation, and define our models. In section 4, we study a common subproblem, and then describe an efficient algorithm for solving each model. In section 5, we conduct a computational study, and identify several relevant managerial insights. Finally, in section 6, we provide a conclusion and suggestions for future research.

2. Literature Review

There are two types of approaches that exist for modeling and solving dynamic pricing problems with limited demand information. For parametric problems, most existing studies use *demand learning approaches*. In such approaches, price experimentation is used to learn about unknown demand parameters and incorporated into price optimization. Such approaches usually require many data points and computational iterations to estimate parameters, and hence may not be practical within a short time horizon. For nonparametric problems, most studies use *robust optimization approaches* that optimize a worst-case performance measure, for example, maximizing the minimum revenue, minimizing the maximum regret, or maximizing the minimum competitive ratio, over all possible demand realizations.

We review only studies of nonparametric problems or robust optimization approaches. Perakis and Sood (2006), Lim and Shanthikumar (2007), Lim et al. (2008) and Cohen et al. (2013) develop robust optimization approaches for dynamic pricing problems. Perakis and Sood (2006) consider a parametric problem where multiple firms sell differentiated products, and propose a robust optimization approach that maximizes the worst-case revenue for each firm. The problems studied by Lim and Shanthikumar (2007), Lim et al. (2008) and Cohen et al. (2013) are all nonparametric. Lim and Shanthikumar (2007) use relative entropy to model *ambiguity* in the demand rate and formulate the problem as a two-player zero-sum stochastic differential game. Lim et al. (2008) use relative entropy to model demand uncertainty for multiple products, and formulate the problem as a worst-case stochastic intensity control problem. Cohen et al. (2013) propose a sampling-based model with limited data.

Besbes and Zeevi (2009, 2012), Eren and Maglaras (2010) and Wang et al. (2014) study nonparametric dynamic pricing problems with demand learning. Besbes and Zeevi (2009) assume that the functional form of demand is unknown but satisfies some standard conditions, and minimize the maximum regret. They propose an algorithm with asymptotically optimal performance as the initial inventory and demand rate become large. Wang et al. (2014) consider the same problem and propose a different algorithm with lower regret. Eren and Maglaras (2010) consider two problems in which arriving customers' willingness-to-pay follows an unknown i.i.d. distribution. The objective is to maximize the worst-case competitive ratio over all such distributions. Besbes and Zeevi (2012) consider a multi-product problem with the objective of maximizing the worst-case competitive ratio. They propose algorithms that are asymptotically optimal as the volume of sales becomes large.

There are important differences between our work and those above. First, in our dynamic model, regret is calculated dynamically, such that the DM's and AD's decisions are interdependent over time. Second, our study is the first analysis of robust nonparametric dynamic pricing problems with interval uncertainty. Third, the algorithms in those existing studies are heavily computational and approximate; whereas, our algorithms quickly generate optimal solutions.

3. Models

We consider a DM who makes pricing decisions for selling a given amount of inventory, I_0 , of a single product over two periods. Let p_t denote the price chosen by the DM in period t , $t = 1, 2$. The product can be offered at discrete prices $p_{t1} < \dots < p_{tm_t}$, where m_t is the number of allowable prices in period t , for $t = 1, 2$. Let $\mathcal{P} = \{p_{11}, \dots, p_{1m_1}\} \times \{p_{21}, \dots, p_{2m_2}\}$ denote the set of all possible prices over the two periods. In each period $t = 1, 2$, the product is subject to a demand $d_t(p_t)$ that depends stochastically on the chosen price p_t . However, the DM does not have complete information about the demand distributions. Instead, for a given price p_t , the DM knows only that

$$l_t(p_t) \leq d_t(p_t) \leq u_t(p_t), \quad t = 1, 2, \quad (1)$$

where $l_t(p_t)$ and $u_t(p_t)$ represent given lower and upper bounds, respectively, on the demand in period t for the given price p_t .

We assume that there is no demand correlation across the two time periods. The stochastic demand in each period is a nonincreasing function of price, i.e.,

$$\begin{aligned} \text{If } p'_t \leq p''_t, \text{ then } d_t(p'_t) \geq d_t(p''_t), l_t(p'_t) \geq l_t(p''_t), \\ \text{and } u_t(p'_t) \geq u_t(p''_t), \text{ for } t = 1, 2. \end{aligned} \quad (2)$$

In order to model the problem faced by the DM using regret, we assume the existence of an AD with the same solution space for prices as the DM. Let $q_t \in \mathcal{P}$ denote the price chosen by the AD in period t , $t = 1, 2$. For a given set of prices (p_1, p_2) chosen by the DM, the AD specifies her own set of prices (q_1, q_2) and also specifies demand curves $(d_1, d_2) \in \mathcal{D}$, where $\mathcal{D}$ is the set of all possible demand curves. We assume that all the demand curves in $\mathcal{D}$ satisfy (2).

Observe that the actual sales in a period is always no more than the available inventory. Hence, for any given price $p_t \in \{p_{t1}, \dots, p_{tm_t}\}$ in period $t = 1, 2$, if $I_0 < l_t(p_t)$, then we may redefine $l_t(p_t) = I_0$ and $u_t(p_t) = I_0$, and if $l_t(p_t) \leq I_0 < u_t(p_t)$, then we may redefine $u_t(p_t) = I_0$. Since the problem with the demand intervals redefined in this way is equivalent to the original problem, we henceforth assume these redefinitions where necessary.

The pricing decisions of the AD are notional constructs for the purposes of comparing the revenue of the DM and the revenue in an ideal, i.e. clairvoyant, situation. Thus, the only prices visible to customers are those specified by the DM. The specifications of prices and demand curves by the AD are made so as to maximize the total difference between the AD's revenue and the DM's revenue over the two periods. In each period t , the regret of the DM, ρ_t , is calculated as the difference between the revenue of the AD and that of the DM. Let $\rho = \rho_1 + \rho_2$ denote the total regret of the DM over the two periods. Since, in any instance, the AD can copy the DM's decisions, the total regret is always nonnegative. However, it is pos-

sible that the regret in the first period, ρ_1 , can be negative. The DM's objective is to minimize ρ over his pricing decisions. The AD's objective is to maximize ρ over her pricing and demand curve decisions.

We consider two models of this problem, as follows.

3.1. Model 1 (dynamic model)

The DM makes his pricing decision contingent on the remaining inventory at the beginning of each period. The sequence of decisions is as follows. At the beginning of $t = 1$, first the DM sets his price p_1 , then the AD specifies the demand realization $d_1(p_1)$ which is revealed to the DM, sets her price q_1 and specifies the demand realization $d_1(q_1)$. Neither q_1 nor $d_1(q_1)$ is revealed to the DM. At the beginning of $t = 2$, given his remaining inventory $I_1 = I_0 - d_1(p_1)$, the DM sets his price p_2 . Knowing this, and her remaining inventory $J_1 = I_0 - d_1(q_1)$, the AD sets her price q_2 and

specifies the demand realizations $d_2(p_2)$ and $d_2(q_2)$, where $d_2(p_2)$ is revealed to the DM, but q_2 and $d_2(q_2)$ are not. Figure 1 illustrates the timeline of the decisions and information flow in Model 1, where a directed arrow from one party (DM or AD) to the other party (AD or DM, respectively) represents that the corresponding decision by the DM or AD is revealed to the AD or DM; a directed arrow from the AD to the timeline represents that the AD's corresponding decision is not revealed to the DM; and revelation of uncertain demand is indicated by a directed arrow from the timeline to the DM.

Given the above definitions, the formulation of the minimax regret problem for this model is as follows:

$$\min_{p_1} \max_{\substack{d_1(p_1) \\ q_1, d_1(q_1)}} \left\{ [q_1 d_1(q_1) - p_1 d_1(p_1)] + \min_{p_2} \max_{\substack{q_2, d_2(q_2) \\ d_2(p_2)}} \{q_2 \min\{J_1, d_2(q_2)\} - p_2 \min\{I_1, d_2(p_2)\}\} \right\}, \quad (3)$$

3.2. Model 2 (static model)

The DM sets his prices for both periods p_1 and p_2 at the beginning of the first period. This enables the AD to make all her decisions in period 1. Thus, Model 2 can be considered as a static model and less restrictive for the AD than Model 1. The AD can make her decisions in an arbitrary sequence or all decisions at the same time. Consequently, this model will deliver a higher regret for the DM than Model 1. Figure 2 illustrates an example for the timeline of the decisions and information flow in Model 2, using the same format as Figure 1.

Figure 1 Timeline of Model 1

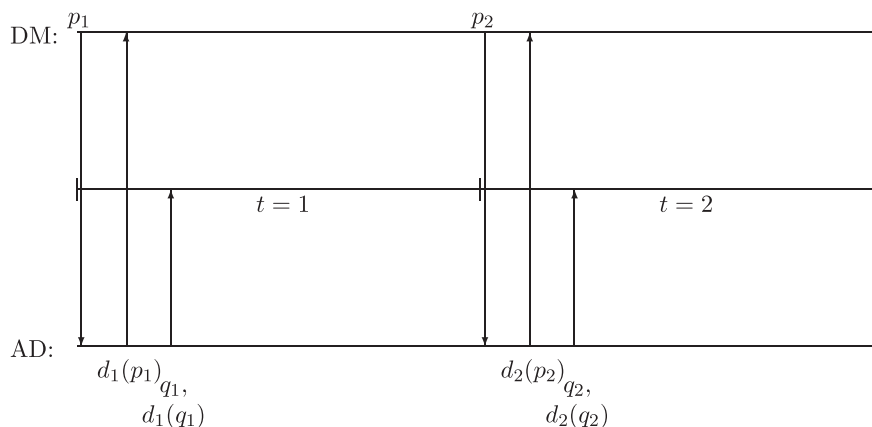
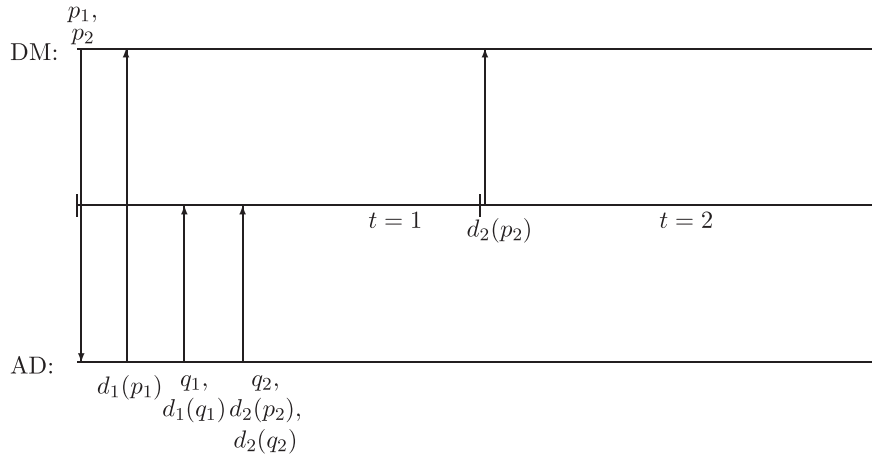


Figure 2 Timeline of Model 2



Model 2 can be formulated as follows:

$$\begin{aligned} \min_{p_1, p_2} \quad & \max_{q_1, q_2,} \quad \{q_1 d_1(q_1) - p_1 d_1(p_1) \\ & d_1(p_1), d_1(q_1), \quad + q_2 \min\{J_1, d_2(q_2)\} \\ & d_2(q_2), d_2(p_2) \quad - p_2 \min\{I_1, d_2(p_2)\}\}, \end{aligned} \quad (4)$$

where $I_1 = I_0 - d_1(p_1)$, $J_1 = I_0 - d_1(q_1)$ and $q_1, d_1(q_1), J_1, q_2$, and $d_2(q_2)$ are not revealed directly to the DM. As in Model 1, the DM can infer the values of these variables by solving specific optimization problems to be discussed below.

4. Solution Methods

In section 4.1, we develop an algorithm to solve a subproblem that is common to both Models 1 and 2. Sections 4.2 and 4.3 provide algorithms for solving Models 1 and 2, respectively.

4.1. Solving a Common Subproblem

In this section, we solve a maximization problem faced by the AD, which appears as a subproblem in the second period in both Models 1 and 2. The formulations (3) and (4) for Models 1 and 2, respectively, contain this subproblem. Supposing that p_2, q_2, I_1 and J_1 are fixed, the AD's remaining problem in the second period in formulation (3), and part of the AD's problem in formulation (4), is to decide the values of $d_2(p_2)$ and $d_2(q_2)$ to maximize the DM's regret in the second period. This problem, denoted by P , is defined as follows:

$$\begin{aligned} Y(p_2, q_2, I_1, J_1) = \quad & \max_{d_2(p_2), d_2(q_2)} \quad \Phi(d_2(p_2), d_2(q_2)) \\ & = q_2 \min\{J_1, d_2(q_2)\} - p_2 \min\{I_1, d_2(p_2)\} \\ \text{s.t. } \quad & d_2(p_2) \leq d_2(q_2), \text{ if } q_2 \leq p_2 \\ & d_2(q_2) \leq d_2(p_2), \text{ if } p_2 \leq q_2 \\ & l_2(p_2) \leq d_2(p_2) \leq u_2(p_2) \\ & l_2(q_2) \leq d_2(q_2) \leq u_2(q_2). \end{aligned} \quad (5)$$

The first two constraints in (5) correspond to the fact that the stochastic demand is a nonincreasing function of price, as in (2).

Given Problem P , Model 1 can be reformulated as:

$$\min_{p_1} \max_{d_1(p_1), q_1, d_1(q_1)} \left\{ \begin{aligned} & [q_1 d_1(q_1) - p_1 d_1(p_1)] \\ & + \min_{p_2} \max_{q_2} Y(p_2, q_2, I_1, J_1) \end{aligned} \right\}. \quad (6)$$

Similarly, Model 2 can be reformulated as:

$$\min_{p_1, p_2} \max_{q_1, q_2, d_1(p_1), d_1(q_1)} \{ [q_1 d_1(q_1) - p_1 d_1(p_1)] + Y(p_2, q_2, I_1, J_1) \}. \quad (7)$$

If $p_2 \geq q_2$, we have from (2) that:

$$l_2(p_2) \leq l_2(q_2) \quad \text{and} \quad u_2(p_2) \leq u_2(q_2). \quad (8)$$

Alternatively, if $p_2 \leq q_2$, we have from (2) that:

$$l_2(q_2) \leq l_2(p_2) \quad \text{and} \quad u_2(q_2) \leq u_2(p_2). \quad (9)$$

We now discuss how to solve problem P , given p_2, q_2, I_1 and J_1 , using several cases. Let $d_2^*(q_2)$ and $d_2^*(p_2)$ denote an optimal solution of problem P .

Case 1: $J_1 \leq l_2(q_2)$. In this case, demand exceeds the inventory of the AD. Hence, we have $\Phi = q_2 J_1 - p_2 \min\{I_1, d_2(p_2)\}$. Choosing $d_2^*(q_2) = l_2(q_2)$ guarantees from (8) or (9) that the smallest possible value for $d_2^*(p_2)$, i.e., $d_2^*(p_2) = l_2(p_2)$, satisfies the first two constraints of (5). Thus, an optimal solution is $d_2^*(q_2) = l_2(q_2)$, $d_2^*(p_2) = l_2(p_2)$, with regret value $Y_1 = q_2 J_1 - p_2 \min\{I_1, l_2(p_2)\}$.

Case 2: $J_1 > l_2(q_2)$. Observe first that if $d_2(q_2) \geq \min\{J_1, u_2(q_2)\}$, then $d_2(q_2)$ can be decreased to $\min\{J_1, u_2(q_2)\}$ and the value of $\Phi(\cdot, \cdot)$ does not decrease.

We distinguish three subcases.

Case 2.A: $p_2 > q_2$. The AD chooses the largest value for $d_2^*(q_2)$ and the smallest possible value for $d_2^*(p_2)$, i.e., $d_2^*(q_2) = \min\{J_1, u_2(q_2)\}$ and $d_2^*(p_2) = l_2(p_2)$. This satisfies the first constraint of (5), since we have

$$d_2^*(p_2) = l_2(p_2) \leq l_2(q_2) \leq \min\{J_1, u_2(q_2)\} = d_2^*(q_2),$$

where the first inequality follows from (8). The regret value is $Y_{2A} = q_2 \min\{J_1, u_2(q_2)\} - p_2 \min\{I_1, l_2(p_2)\}$.

Case 2.B: $p_2 \leq q_2$ and $\min\{J_1, u_2(q_2)\} \leq l_2(p_2)$. We distinguish three subcases. First, suppose $l_2(p_2) > u_2(q_2)$. Then, from (2), $p_2 < q_2$. Then, the AD chooses the largest feasible value for $d_2^*(q_2)$ and the smallest feasible value for $d_2^*(p_2)$, i.e., $d_2^*(q_2) = \min\{J_1, u_2(q_2)\}$ and $d_2^*(p_2) = l_2(p_2)$. The regret value is $Y_{2B} = q_2 \min\{J_1, u_2(q_2)\} - p_2 \min\{I_1, l_2(p_2)\}$.

In the second subcase, suppose $J_1 \leq l_2(p_2) \leq u_2(q_2)$. Again, the AD chooses the smallest feasible value for $d_2^*(p_2)$, i.e. $d_2^*(p_2) = l_2(p_2)$. For all feasible values of $d_2(q_2)$ greater than or equal to J_1 , the regret value is a constant. Hence, the AD chooses $d_2^*(q_2) = l_2(p_2)$. This choice also guarantees feasibility for the case $p_2 = q_2$. The regret value is $q_2 \min\{J_1, l_2(p_2)\} - p_2 \min\{I_1, l_2(p_2)\}$, which is equal to Y_{2B} since $J_1 \leq l_2(p_2) \leq u_2(q_2)$.

In the third subcase, $J_1 > l_2(p_2) = u_2(q_2)$. In this case, it can be seen that $d_2^*(p_2) = d_2^*(q_2) = l_2(p_2) = u_2(q_2)$, and the regret value is again equal to $Y_{2B} = q_2 \min\{J_1, u_2(q_2)\} - p_2 \min\{I_1, l_2(p_2)\}$.

Case 2.C: $p_2 \leq q_2$ and $\min\{J_1, u_2(q_2)\} > l_2(p_2)$. Recall the Case 2 condition $J_1 > l_2(q_2)$. Since increasing the sales of both parties increases the regret, the AD chooses $d_2^*(q_2) = d_2^*(p_2) = \min\{J_1, u_2(q_2)\}$. This solution is feasible for problem P , since we have

$$l_2(p_2) < \min\{J_1, u_2(q_2)\} = d_2^*(p_2) \leq u_2(q_2) \leq u_2(p_2),$$

where the last inequality follows from (9). The regret value is $Y_{2C} = q_2 \min\{J_1, u_2(q_2)\} - p_2 \min\{I_1, J_1, u_2(q_2)\}$.

Table 1 summarizes the various cases discussed above, and shows the corresponding optimal values of $d_2^*(q_2)$ and $d_2^*(p_2)$, and the maximum regret value Y .

4.2. Solving Model 1

In this section, we describe an algorithm for solving Model 1, i.e. problem (6). The DM constructs a matrix of regret values for all possible pairs (p_1, q_1) of first period prices, and performs a $\min_{p_1} - \max_{q_1}$ optimization over this matrix.

In order to see how those regret values are obtained, we consider the AD's $\max_{d_1(p_1), d_1(q_1)} \{\cdot\}$ problem, for given (p_1, q_1) values. This problem requires consideration of the total two-period regret. We first consider the second period problem for given values of I_1, J_1 , i.e.,

$$z(I_1, J_1) = \min_{p_2} \max_{q_2} Y(p_2, q_2, I_1, J_1), \quad (10)$$

where $Y(p_2, q_2, I_1, J_1)$ is given in Table 1. To solve problem (10) for given values of I_1, J_1 , the DM constructs a matrix of regret values for all possible pairs (p_2, q_2) of second period prices, and performs a $\min_{p_2} - \max_{q_2}$ optimization over this matrix.

We now characterize the structure of $z(I_1, J_1)$. We divide I_1 and J_1 into at most $2m_2 + 1$ intervals as follows. Let $K \leq 2m_2 + 2$ be the number of distinct values in the set, $\{l_2(p_{2i}), u_2(p_{2i}) | i = 1, \dots, m_2\} \cup \{0, I_0\}$. Sort these distinct values in ascending order and denote them by a_j , for $j = 1, \dots, K$. Divide the domain of I_1 and J_1 into $K - 1$ intervals, $[a_j, a_{j+1}]$, for $j = 1, \dots, K - 1$. Using these intervals, we divide the joint domain of (I_1, J_1) into $L = (K - 1)^2 + (K - 1)$ subdomains, denoted by $x_l(I_1, J_1)$, for $l = 1, \dots, L$, as follows. Each subdomain $x_l(I_1, J_1)$ is specified by $I_1 \in [a_j, a_{j+1}]$ and $J_1 \in [a_h, a_{h+1}]$ for some $j \neq h$, or $I_1, J_1 \in [a_j, a_{j+1}]$ for some j , with a constraint $I_1 \geq J_1$ or $I_1 \leq J_1$. It follows that there are a total of $(K - 1)(K - 2)$ domains $x_l(I_1, J_1)$ where I_1 and J_1 belong to different intervals, and $2(K - 1)$ domains $x_l(I_1, J_1)$, where I_1 and J_1 belong to the same interval but with the constraint $I_1 \geq J_1$ or $I_1 \leq J_1$. Hence, there is a total of L domains $x_l(I_1, J_1)$.

For any given pair of p_2 and q_2 , from Table 1, over a given domain $x_l(I_1, J_1)$, the values of $d_2^*(q_2)$, $d_2^*(p_2)$ and Y are all linear. We thus have the following result.

Table 1 Values of $d_2^*(q_2)$, $d_2^*(p_2)$ and Y under Various Cases

Case	Conditions	$d_2^*(q_2)$	$d_2^*(p_2)$	$Y(p_2, q_2, I_1, J_1)$
1	$J_1 \leq l_2(q_2)$	$l_2(q_2)$	$l_2(p_2)$	$Y_1 = q_2 J_1 - p_2 \min\{I_1, l_2(p_2)\}$
2.A	$J_1 > l_2(q_2); p_2 > q_2$	$\min\{J_1, u_2(q_2)\}$	$l_2(p_2)$	$Y_{2A} = q_2 \min\{J_1, u_2(q_2)\} - p_2 \min\{I_1, l_2(p_2)\}$
2.B1	$J_1 > l_2(q_2); p_2 \leq q_2; u_2(q_2) < l_2(p_2)$	$\min\{J_1, u_2(q_2)\}$	$l_2(p_2)$	$Y_{2B} = q_2 \min\{J_1, u_2(q_2)\} - p_2 \min\{I_1, l_2(p_2)\}$
2.B2	$J_1 > l_2(q_2); p_2 \leq q_2; J_1 \leq l_2(p_2) \leq u_2(q_2)$	$l_2(p_2)$	$l_2(p_2)$	$Y_{2B} = q_2 \min\{J_1, u_2(q_2)\} - p_2 \min\{I_1, l_2(p_2)\}$
2.B3	$J_1 > l_2(q_2); p_2 \leq q_2; J_1 > l_2(p_2) = u_2(q_2)$	$u_2(q_2)$	$l_2(p_2)$	$Y_{2B} = q_2 \min\{J_1, u_2(q_2)\} - p_2 \min\{I_1, l_2(p_2)\}$
2.C	$p_2 \leq q_2; \min\{J_1, u_2(q_2)\} > l_2(p_2)$	$\min\{J_1, u_2(q_2)\}$	$\min\{J_1, u_2(q_2)\}$	$Y_{2C} = q_2 \min\{J_1, u_2(q_2)\} - p_2 \min\{I_1, J_1, u_2(q_2)\}$

LEMMA 1. Over any given domain $x_l(I_1, J_1)$, for $l = 1, \dots, L$, the optimal solutions $d_2^*(q_2)$, $d_2^*(p_2)$ and the corresponding maximum regret Y are all linear functions of I_1 and J_1 .

Now, let $\phi(p_2) = \max_{q_2} Y(p_2, q_2, I_1, J_1)$. Over any given domain $x_l(I_1, J_1)$, $\phi(p_2)$ is the upper envelope of the functions $Y(p_2, p_{2i}, I_1, J_1)$, for $i = 1, \dots, m_2$. From Lemma 1, the functions $Y(p_2, p_{2i}, I_1, J_1)$ are linear. Thus, $\phi(p_2)$ is a piecewise linear function of I_1 and J_1 with at most m_2 pieces over $x_l(I_1, J_1)$. Further, let $z(I_1, J_1) = \min_{p_2} \phi(p_2)$. Then, $z(I_1, J_1)$ is the lower envelope of at most m_2^2 pieces, and is hence a piecewise linear function of I_1 and J_1 with at the most m_2^2 pieces over $x_l(I_1, J_1)$, where the planes consist of at most m_2 segments of the piecewise linear function $\phi(p_{2i})$, for $i = 1, \dots, m_2$. We therefore divide each domain $x_l(I_1, J_1)$ into at most m_2^2 sub-subdomains, each of which is described by a set of linear inequalities for I_1 and J_1 , such that $z(I_1, J_1)$ is a linear function of I_1 and J_1 over each such sub-subdomain.

In each of the L domains $x_l(I_1, J_1)$, there are at most m_2^2 sub-subdomains over which $z(I_1, J_1)$ is a linear function of I_1 and J_1 . Hence, the feasible space of I_1 and J_1 can be divided into at most $Lm_2^2 = O(m_2^4)$ sub-subdomains, over each of which $z(I_1, J_1)$ is a linear function of I_1 and J_1 . Let $H = O(m_2^4)$ denote the total number of such sub-subdomains, and $y_h(I_1, J_1)$, for $h = 1, \dots, H$, denote these domains. The following lemma summarizes the above discussion.

LEMMA 2. Over any given domain $y_h(I_1, J_1)$, for $h = 1, \dots, H$, the corresponding $z(I_1, J_1)$ is a linear function of I_1 and J_1 .

For any given pair of (p_1, q_1) , the AD's first period problem of specifying demand realizations to maximize regret is equivalent to finding a domain $y_h(I_1, J_1)$ for some h , such that the objective value of the following two variable linear program is maximum:

$$\max_{d_1(q_1), d_1(p_1)} [q_1 d_1(q_1) - p_1 d_1(p_1)] + z(I_0 - d_1(p_1), I_0 - d_1(q_1)) \quad (11)$$

$$\text{s.t. } l_1(p_1) \leq d_1(p_1) \leq u_1(p_1) \quad (12)$$

$$l_1(q_1) \leq d_1(q_1) \leq u_1(q_1) \quad (13)$$

$$d_1(p_1) \leq d_1(q_1), \quad \text{if } p_1 \geq q_1 \quad (14)$$

$$d_1(p_1) \geq d_1(q_1), \quad \text{if } p_1 \leq q_1 \quad (15)$$

$I_1 = I_0 - d_1(p_1)$ and $J_1 = I_0 - d_1(q_1)$ satisfy all the linear inequalities that describe domain $y_h(I_1, J_1)$.
 (16)

For each given pair (p_1, q_1) , we solve the above linear program for each $y_h(I_1, J_1)$, for $h = 1, \dots, H$, and let $Z(p_1, q_1)$ denote the maximum objective value over all the H domains. This gives a matrix of regret values $Z(p_1, q_1)$. Problem (6) is then solved by performing a min max optimization over this matrix.

The following result summarizes the discussion in this section.

THEOREM 1. An optimal solution for Model 1 can be found in $O(m_1^2 m_2^4)$ time.

PROOF. The optimality of the algorithm follows directly from the above discussions. To solve Model 1, we solve the problem $\min_{p_1} \max_{q_1} Z(p_1, q_1)$ by enumerating all possible values of p_1 and q_1 . Hence, it requires $O(m_1^2 V)$ time to solve Model 1, where V is the time needed to calculate $Z(p_1, q_1)$ for given p_1 and q_1 . We now show that $V = O(m_2^4)$. Within each of the $O(m_2^4)$ sub-subdomains $y_h(I_1, J_1)$ described, the linearity of $z(I_1, J_1)$ is established by Lemma 2. Furthermore, following the discussion prior to Lemma 2, for any given p_1 and q_1 , the linear functions $z(I_1, J_1)$ over all domains $y_h(I_1, J_1)$ can be found in $O(m_2^4)$ time. For any given p_1 and q_1 , the linear program (11)–(16) is solved $O(m_2^4)$ times, once for each domain $y_h(I_1, J_1)$. It is known that two-variable linear programs can be solved in time linear in the number of constraints (Megiddo 1984). Also, the number of inequalities in Equation (16) is a constant. Thus, for given p_1 and q_1 , it takes $O(m_2^4)$ time to solve all the linear programs required. $\square$

In the online Appendix S1, we provide a small numerical example to illustrate the step-by-step procedures of the algorithm described above.

4.3. Solving Model 2

In this section, we outline an algorithm for solving Model 2, i.e., problem (7). For given p_1, p_2, q_1 and q_2 , the following problem, denoted by P_1 , represents the optimal choice of $d_1(p_1)$ and $d_1(q_1)$ by the AD. The value of function $Y(\cdot)$ is given in Table 1.

$$\begin{aligned} \tilde{Z}(p_1, p_2, q_1, q_2) &= \max_{d_1(p_1), d_1(q_1)} \Psi(d_1(p_1), d_1(q_1)) \\ &= q_1 d_1(q_1) - p_1 d_1(p_1) + Y(p_2, q_2, I_0 - d_1(p_1), I_0 - d_1(q_1)) \\ \text{s.t. } d_1(q_1) &\geq d_1(p_1), \quad \text{if } q_1 \leq p_1 \end{aligned} \quad (17)$$

$$d_1(q_1) \leq d_1(p_1), \text{ if } p_1 \leq q_1 \quad (18)$$

$$l_1(p_1) \leq d_1(p_1) \leq u_1(p_1) \quad (19)$$

$$l_1(q_1) \leq d_1(q_1) \leq u_1(q_1). \quad (20)$$

We outline below how problem P_1 can be solved efficiently. A detailed analysis is given in Appendix A. Let $d_1^*(p_1)$, $d_1^*(q_1)$ denote not necessarily unique values of $d_1(p_1)$ and $d_1(q_1)$ that solve problem P_1 optimally. The corresponding value $\Psi(d_1^*(p_1), d_1^*(q_1))$ depends on the parameters p_1 , q_1 , p_2 and q_2 , and is denoted by $\tilde{Z} \equiv \tilde{Z}(p_1, p_2, q_1, q_2)$. Thus,

$$\tilde{Z} = q_1 d_1^*(q_1) - p_1 d_1^*(p_1) + Y(p_2, q_2, I_0 - d_1^*(p_1), I_0 - d_1^*(q_1)).$$

Then, from (7), it follows that Model 2 can be written as:

$$\min_{p_1, p_2} \max_{q_1, q_2} \tilde{Z}(p_1, p_2, q_1, q_2). \quad (21)$$

Model 2 is then solved using (21) by enumerating all possible values of p_1 , p_2 , q_1 and q_2 and performing a $\min_{p_1, p_2} \max_{q_1, q_2}$ optimization.

To solve problem P_1 , we first note that the function $\Psi(\cdot)$ may have different forms under different cases in Table 1. In Cases 1, 2.A and 2.B of Table 1, the function $\Psi(\cdot)$ can be written as

$$\begin{aligned} \Psi(d_1(p_1), d_1(q_1)) &\equiv \Psi_a = q_1 d_1(q_1) - p_1 d_1(p_1) \\ &\quad + q_2 \min\{I_0 - d_1(q_1), u_2(q_2)\} \\ &\quad - p_2 \min\{I_0 - d_1(p_1), l_2(p_2)\}. \end{aligned} \quad (22)$$

In Case 2.C, $\Psi(\cdot)$ can be written as

$$\begin{aligned} \Psi(d_1(p_1), d_1(q_1)) &\equiv \Psi_b = q_1 d_1(q_1) - p_1 d_1(p_1) \\ &\quad + q_2 \min\{I_0 - d_1(q_1), u_2(q_2)\} \\ &\quad - p_2 \min\{I_0 - d_1(p_1), \\ &\quad I_0 - d_1(q_1), u_2(q_2)\}. \end{aligned} \quad (23)$$

If $p_1 \geq q_1$, the following constraints must be satisfied, from (2):

$$d_1(q_1) \geq d_1(p_1), u_1(q_1) \geq u_1(p_1), l_1(q_1) \geq l_1(p_1). \quad (24)$$

Similarly, if $p_1 \leq q_1$, the following constraints must be satisfied, from (2):

$$d_1(p_1) \geq d_1(q_1), u_1(p_1) \geq u_1(q_1), l_1(p_1) \geq l_1(q_1). \quad (25)$$

In Appendix A, we show that for any given values of the parameters p_1 , q_1 , p_2 , q_2 , we can partition the domain of $(d_1(q_1), d_1(p_1))$ into a constant number of subdomains such that within each such subdomain, the function Ψ is a linear function of $d_1(p_1)$ and $d_1(q_1)$,

and problem P_1 becomes a two-variable linear program, which can be solved in a constant time. Since each parameter p_1 and q_1 can attain m_1 different values and each parameter p_2 and q_2 can attain m_2 different values, we have the following result.

THEOREM 2. *An optimal solution for Model 2 can be found in $O(m_1^2 m_2^2)$ time.*

In the online Appendix S1, we provide a small numerical example with a brief discussion about how the example can be solved by the algorithm described above.

5. Computational Results and Insights

In this section, we conduct computational experiments to gain managerial insights about dynamic pricing decisions for our problems. In section 5.1, we describe how test problems are generated. In section 5.2, we consider the sensitivity of the DM's pricing decisions and the maximum regret to several problem parameters. In section 5.3, we compare the performance of our approach of minimizing the maximum regret with several other approaches as benchmarks.

5.1. Parameter Values

The main problem parameters used in our experiments are generated as follows:

- Number of allowable prices: $m_1 = m_2 = 10$.
- Allowable prices: $p_{t1} = \$50$, $p_{ti+1} = p_{ti} + \lfloor 0.1 p_{ti} \rfloor$, for $i = 1, \dots, m_1$, $t = 1, 2$. Therefore, in both periods, the 10 allowable prices are \$50, \$55, \$60, \$66, \$72, \$79, \$86, \$94, \$103, \$113. Notice that the difference between any two neighboring prices is approximately 10%.
- Demand intervals $[l_t(p_t), u_t(p_t)]$, for $t = 1, 2$: For a given allowable price p , we first generate a base demand, following a constant elasticity model: $d(p) = 50(p/p_{\max})^{-1.5}$, where $p_{\max}$ is the maximum allowable price. A parameter $spread_t$ is used to control the length of the demand interval in period t . For a given price p , we define $l_t(p) = \lfloor d(p) - spread_t \times d(p) \rfloor$, and $u_t(p) = \lfloor d(p) + spread_t \times d(p) \rfloor$, for $t = 1, 2$, where $spread_t \in \{0.05, 0.1, 0.2, 0.4\}$. These values of $spread_t$ imply that the width of the demand interval is 10%, 20%, 40%, and 80% of the base demand, respectively. We consider all sixteen possible combinations of $(spread_1, spread_2)$, which together cover a variety of demand uncertainty levels in the two periods.
- Initial inventory I_0 : We let $I_0 = \alpha(I_0)D$, where $\alpha(I_0) \in \{0.5, 1, 1.5\}$, $D = D_1 + D_2$ and D_t is the midpoint of the demand interval corresponding

to the median allowable price in period t , i.e. $p_{t, \lceil m_t/2 \rceil}$. The three cases of $\alpha(I_0)$ represent small, medium, and large I_0 values relative to the total demand, respectively.

The algorithms developed in sections 4.2 and 4.3 can solve every test problem with the above specified parameters in less than one minute. Hence, we do not report specific computational times for these algorithms below.

5.2. Sensitivity Analysis

We study how the optimal prices, maximum total regret, and regret in the first period change with the level of demand uncertainty and the initial inventory, I_0 . Our computational results are given in a table in Appendix B. We summarize these results as follows. Let p_t^* be the optimal price used in period t in the optimal solution, for $t = 1, 2$. Our results show that, given initial inventory I_0 , the optimal prices generated by both of our models are insensitive to the level of demand uncertainty. More specifically, for each level of I_0 , p_1^* falls in a range of 10%, that is, two consecutive price points, across all uncertainty levels, with very few exceptions. Similarly, p_2^* falls in a range of 20%, that is, three consecutive price points, in most cases. Hence, even if the demand uncertainty is not accurately estimated, the p_1^* solution generated is typically still optimal or near-optimal. Since the p_2^* solution depends on the remaining inventory in the beginning of the second period, its value may vary with the actual demand uncertainty in the first period. The p_2^* solution reported in the table is the solution that achieves the minimum maximum regret when the demand uncertainty in the first period is accurately estimated. That is, the actual demand realization in the first period is always taken from the demand interval generated based on the corresponding *spread*₁ value in the second column of the table. Thus, the p_2^* results shown in the table mean that the p_2^* value generated is also optimal or near-optimal if the demand uncertainty in the first period is accurately estimated.

However, our results show that the minimax regret value generated by both of our models increases significantly with the level of demand uncertainty as well as the initial inventory. We prove below that it is always true that the minimax regret value generated by both of our models is nondecreasing in the level of demand uncertainty. However, it is an open question whether it is always true that the minimax regret value generated by one or both of our models is nondecreasing in the initial inventory. Intuitively, this result holds because, while a higher initial inventory provides more options for both the DM and the AD, it benefits the AD more since she has more information than the DM before she makes her decisions.

THEOREM 3. *The minimax regret value of our models is nondecreasing in the widths of the demand intervals.*

PROOF. We show the result for Model 1. Suppose we have two instances, I and I' , of Model 1 with the same set of allowable prices but the demand intervals under instance I' are wider, i.e. $l'(p_t) \leq l(p_t)$ and $u(p_t) \leq u'(p_t)$, for any allowable price p_t , in each period $t = 1, 2$. Thus, the demand space under instance I is a subset of the demand space under instance I' .

For instance I , for any price p_1 chosen by the DM in period 1, define $F(p_1)$ to be the maximum regret value for the DM. $F(p_1)$ can be written as the following optimization problem solved by the AD:

$$F(p_1) = \max_{q_1, d_1(p_1), d_1(q_1)} q_1 d_1(q_1) - p_1 d_1(p_1) + z(I_1, J_1),$$

where $z(I_1, J_1) = \min_{p_2} g(p_2, I_1, J_1)$, and $g(p_2, I_1, J_1)$ is the second period regret for the DM, given that the DM chooses p_2 as his second period price, I_1 is the remaining inventory of the DM and J_1 is the remaining inventory of the AD. We have

$$g(p_2, I_1, J_1) = \max_{q_2, d_2(p_2), d_2(q_2)} q_2 \min(J_1, d_2(q_2)) - p_2 \min(I_1, d_2(p_2)).$$

For instance I' , we define $F'(p_1)$, $z'(I_1, J_1)$ and $g'(p_2, I_1, J_1)$ in exactly the same way, except that $d_1(p_1)$, $d_1(q_1)$, $d_2(p_2)$ and $d_2(q_2)$ are now taken from the wider demand intervals under instance I' .

First, it can be easily seen that for any given p_2 , I_1 and J_1 , $g'(p_2, I_1, J_1) \geq g(p_2, I_1, J_1)$. Thus, for any given I_1 , and J_1 , $z'(I_1, J_1) \geq z(I_1, J_1)$. This implies that, for any given p_1 , q_1 , $d_1(p_1)$, and $d_1(q_1)$, the objective value in the maximization problem for $F'(p_1)$ is greater than or equal to that in the maximization problem for $F(p_1)$. Since the former maximization problem has a domain larger than that of the latter maximization problem, we have $F'(p_1) \geq F(p_1)$, for any given p_1 . Thus, $\min_{p_1} F'(p_1) \geq \min_{p_1} F(p_1)$, that is, the total regret of instance I' is greater than or equal to that of instance I . A similar proof can be given for Model 2. $\square$

The result in Theorem 3 is the only conclusion we can draw about the relationship between the minimax regret and the widths of the demand intervals. From our computational results, (i) in general, the minimax regret value is neither a convex nor a concave function of the widths of the demand intervals in any period, and (ii) the widths of the demand intervals in both periods play an equally significant role in determining the minimax regret value.

Another observation we can make from our computational results is that the optimal prices in both

periods generated by both of our models decrease substantially when the initial inventory I_0 increases. It is thus natural to investigate whether it is always true that the optimal prices for our models are nonincreasing in the given initial inventory. Gallego and van Ryzin (1994) show that such a relationship between the optimal price and the inventory level always exists in the traditional dynamic pricing problem where the allowable prices are continuous (i.e., any value in a given interval is an allowable price) and the objective is to maximize the total expected revenue. However, we can find a counterexample, given below, to show that such a result does not always hold for our models even when there is no demand uncertainty, that is, the lower and upper bounds of the price intervals are identical.

EXAMPLE. Given two cases of initial inventory $I_0 = 10$ or 20. For both period 1 and 2, there are two allowable prices $r_1 = 2$, $r_2 = 6$. The demand for each price is deterministic as follows: in period 1, $d_1(r_1) = 6$, $d_1(r_2) = 1$; in period 2, $d_2(r_1) = 19$, $d_2(r_2) = 4$.

Due to the deterministic demand, for our models, the DM and the AD independently choose the same optimal prices, and the optimal objective value is equal to zero. Consequently, the problem of the DM (as well as that of the AD) is to find optimal prices to maximize the total revenue, that is,

$$\max_{p_1, p_2} \Phi(p_1, p_2) = p_1 d_1(p_1) + p_2 \min\{I_0 - d_1(p_1), d_2(p_2)\}. \quad (26)$$

Consider first $I_0 = 10$. There are a total of 4 combinations for (p_1, p_2) , $\{(2,2), (2,6), (6,2), (6,6)\}$. For each given combination of (p_1, p_2) , we compute the objective function as follows.

Case 1: $(p_1, p_2) = (2, 2)$. In this case, $\Phi(p_1, p_2) = 2 \times 6 + 2 \times \min\{10 - 6, 19\} = 20$.

Case 2: $(p_1, p_2) = (2, 6)$. In this case, $\Phi(p_1, p_2) = 2 \times 6 + 6 \times \min\{10 - 6, 4\} = 36$.

Case 3: $(p_1, p_2) = (6, 2)$. In this case, $\Phi(p_1, p_2) = 6 \times 1 + 2 \times \min\{10 - 1, 19\} = 24$.

Case 4: $(p_1, p_2) = (6, 6)$. In this case, $\Phi(p_1, p_2) = 6 \times 1 + 6 \times \min\{10 - 1, 4\} = 30$.

Thus, the optimal solution for the DM is $(2, 6)$.

Now consider $I_0 = 20$. We have the following results for $\Phi(p_1, p_2)$.

Case 1: $(p_1, p_2) = (2, 2)$. In this case, $\Phi(p_1, p_2) = 2 \times 6 + 2 \times \min\{20 - 6, 19\} = 40$.

Case 2: $(p_1, p_2) = (2, 6)$. In this case, $\Phi(p_1, p_2) = 2 \times 6 + 6 \times \min\{20 - 6, 4\} = 36$.

Case 3: $(p_1, p_2) = (6, 2)$. In this case, $\Phi(p_1, p_2) = 6 \times 1 + 2 \times \min\{20 - 1, 19\} = 44$.

Case 4: $(p_1, p_2) = (6, 6)$. In this case, $\Phi(p_1, p_2) = 6 \times 1 + 6 \times \min\{20 - 1, 4\} = 30$.

Thus, for $I_0 = 20$, the optimal solution for the DM is $(6, 2)$. This shows that the optimal price in the first period is actually higher when the inventory is higher.

Finally, we compare our two models. Our computational results show that in most cases where the minimax regret value of Model 2 is nonzero, Model 2 gives a higher minimax regret value than Model 1. As we discuss below in section 5.3, the minimax regret value of Model 2 is typically about 5% to 9% higher than that of Model 1. This is intuitive because Model 2 is a static model where the DM sets the prices for both periods at the beginning of the first period, whereas in Model 1 the DM sets the price for the second period after he sees the demand realization in the first period. Even in most of the cases for which both models have the same optimal prices, the minimax regret value of Model 2 is higher than that of Model 1. This is because in Model 2, the AD sees the DM's prices for both periods at the beginning of the first period and may choose a different demand for the first period than she would for the same period in the case of Model 1.

5.3. Performance of the Minimax Approaches

In this section, we test the robustness of our minimax approaches, Models 1 and 2, by comparing them to three benchmark approaches commonly used in practice based on three commonly used performance measures. In section 5.3.1, we describe how these three approaches can be solved. In section 5.3.2, we provide our results and identify some useful insights.

5.3.1. Benchmarks. When facing demand uncertainty, a commonly used approach in practice is to replace uncertain demand by a deterministic estimate which converts the problem into a deterministic one. For our problem, this can be accomplished by replacing each demand interval by a single point in the interval. We name this procedure the *point estimate approach*. Another commonly used approach when facing demand uncertainty is to optimize a given performance measure under a worst-case scenario, which in our problem context is to maximize the worst-case, i.e., minimum, revenue. We call this the *maximin revenue approach*. Finally, if the probability distribution of the uncertain demand at each allowable price is given, then a commonly used approach, which we call the *maximum expected revenue approach*, is to maximize the total expected revenue by considering the given probability distribution. We now describe how an optimal solution can be found under each of these approaches.

5.3.1.1. Point Estimate Approach: In this approach, we maximize total revenue given that the demand at

each price point is assumed to be known with certainty and equal to the center point of the demand interval. The problem is solved by enumerating all possible pairs of p_1 and p_2 , and taking the solution that maximizes the total revenue.

5.3.1.2. Maximin Revenue Approach: This approach can be formulated as follows:

$$\max_{p_1} \{ \min_{d_1(p_1)} \{ p_1 d_1(p_1) + \max_{p_2} \min_{d_2(p_2)} \{ I_1, d_2(p_2) \} \} \}. \quad (27)$$

Consider the second period problem. Given p_2 , the AD's optimal value of $d_2(p_2)$ is always equal to the minimum possible, i.e., $l_2(p_2)$, regardless of the value of I_1 . Thus, problem (27) can be reformulated as:

$$\max_{p_1} \{ \min_{d_1(p_1)} \{ p_1 d_1(p_1) + \max_{p_2} p_2 \min(I_0 - d_1(p_1), l_2(p_2)) \} \}. \quad (28)$$

Problem (28) is solved as follows. Enumerate every possible value of p_1 . Given p_1 , the inner minimization problem is converted into a two-variable linear program under each of many cases of the domain of $d_1(p_1)$. The domain of $d_1(p_1)$ is divided into $m_2 + 2$ distinct intervals as follows: $[0, I_0 - l_2(p_{2m_2})]$, $[I_0 - l_2(p_{2m_2}), I_0 - l_2(p_{2, m_2-1})]$, $\dots$, $[I_0 - l_2(p_{22}), I_0 - l_2(p_{21})]$, $[I_0 - l_2(p_{21}), I_0]$. When $d_1(p_1)$ is restricted to any of these intervals, say, $[a, b]$, then the inner minimization problem for a given p_1 can be formulated as the following linear program:

$$\begin{aligned} & \min_{d_1(p_1)} p_1 d_1(p_1) + z \\ \text{s.t. } & z \geq p_{2i}(I_0 - d_1(p_1)), \quad \text{if } d_1(p_1) \in [a, b] \text{ implies} \\ & \quad I_0 - l_2(p_{2i}) \leq d_1(p_1), \\ & z \geq p_{2i} l_2(p_{2i}), \quad \text{otherwise, } \forall i = 1, \dots, m_2 \\ & a \leq d_1(p_1) \leq b. \end{aligned}$$

We solve the above two-variable linear program for each possible interval of $d_1(p_1)$, and identify the solution with the minimum objective value as the solution of the overall minimization problem for given p_1 . The value of p_1 is chosen to maximize this revenue.

5.3.1.3. Maximum Expected Revenue Approach: The DM assumes that for each allowable price and time period, there is a known probability distribution of the demand over the given demand interval, and maximizes the total expected revenue. This problem is solved as follows. Consider the second period problem for given I_1 , which is $\max_{p_2} p_2 E[\min(I_1, d_2(p_2))]$. To solve this problem, we enumerate over p_2 . For a given value of p_2 , $E[\min(I_1, d_2(p_2))]$ is calculated as a function of I_1 . Thus, $\max_{p_2} p_2 E[\min(I_1, d_2(p_2))]$ is a function of I_1 . We denote this function by $f(I_1)$. Now consider the first period problem, $\max_{p_1} p_1 E[d_1(p_1)$

$+ f(I_1)]$, which is solved by enumerating p_1 , and for each p_1 calculating the expected value $E[d_1(p_1) + f(I_1)]$ given the probability distribution of $d_1(p_1)$, and taking the maximum value over p_1 .

5.3.2. Results. We first compare our two models with the three benchmark approaches described above, based on three performance measures: (i) maximum regret, (ii) average relative regret, and (iii) total expected revenue if probability distributions of the uncertain demand are known.

Every test problem configuration is determined by specific values of $spread_1$, $spread_2$, and $\alpha(I_0)$, as discussed in section 5.1. For each configuration, we generate one million problem instances following a uniform distribution over each demand interval, where each instance represents a specific demand realization for every allowable price in each period. At the start of the first period, the DM does not have any knowledge of the demand realizations to come. Thus, under any solution approach, the DM uses the optimal price p_1 derived for this approach without assuming any knowledge of the demand realization for the first period. Since Model 2 is static, under this approach, the DM uses the optimal price p_2 derived together with p_1 prior to the beginning of the first period, for the second period. Under any other solution approach, the DM finds the optimal price for the second period p_2 by re-solving the second period problem, given the actual demand realization in the first period. Under each solution approach, for each of the 1 million instances of a test problem, we calculate the maximum possible revenue for this instance by assuming knowledge of the demand realizations before decisions are made. We define the *absolute regret* as the difference between the maximum possible revenue and the total revenue obtained by the DM for this instance, and the *relative regret* as the percentage decrease of the revenue obtained by the DM compared to the maximum possible revenue. For a test problem configuration, the *maximum regret*, the *average relative regret*, and the *total expected revenue* are then the maximum absolute regret, the average relative regret, and the average total revenue over the one million instances, respectively.

It is expected that when the number of instances tested is sufficiently large, among the five approaches including our two models and the three benchmark approaches, Model 1 should achieve the lowest maximum regret, whereas the maximum expected revenue approach should achieve the maximum total expected revenue.

Our computational results are reported in Table 2. The reported maximum regret performance measure represents the percentage increase of the

maximum regret of each other approach compared to the minimax regret value of Model 1. The reported average relative regret performance measure represents the average relative regret of each approach. Finally, the reported total expected revenue performance measure represents the average percentage decrease of the total expected revenue of each other approach compared to that of the maximum expected revenue approach. The five approaches, Model 1, Model 2, Point Estimate, Maximin Revenue, and Maximum Expected Revenue are denoted by M1, M2, PE, MM, and ME, respectively. Based on these results, we have the following observations:

- a. Model 1 outperforms every other approach on the maximum regret and average relative regret performance measures. It generates the smallest maximum regret in every problem configuration, a much smaller maximum regret in most problem configurations, and a smaller average relative regret in most problem configurations. In addition, although Model 1 does not require knowledge of demand probability distributions, it generates a total expected revenue that is very close to that of the maximum expected revenue approach which requires knowledge of demand probability distributions.
- b. Model 2 is outperformed by Model 1 on every performance measure tested in almost every problem configuration. In fact, Model 2, because of its static nature, does not perform well on average relative regret and total expected revenue performance measures. This shows the benefit of dynamic decision making. Further, when the initial inventory is not small, all the three benchmark approaches outperform Model 2 on these performance measures.
- c. The point estimate approach outperforms Model 1, Model 2 and the maximin revenue approach on the total expected revenue performance measure. However, compared to Model 1, it generates significantly higher maximum regret and average relative regret values.

We further compare Model 1 and the point estimate approach on several measures of risk and variability as follows: (i) the standard deviation of the revenue (STD), (ii) the worst-case revenue (WCR), (iii) the maximum revenue target that is achievable with probability of .99, that is, the 1st percentile of the revenue (99R), (iv) the expected revenue for instances in the 1st percentile (99ER), (v) and (vi) as (iii) and (iv) but with probability .95 (95R and 95ER, respectively). Hence, STD measures the variability or robustness of a solution approach, whereas the other measures reflect the worst

revenue realization, worst 1% of possible revenue realizations, and worst 5% of possible revenue realizations, respectively, thus collectively measuring downside risk. We note that the measures 95R and 99R are similar to value at risk (VaR), and the measures 95ER and 99ER are similar to conditional value at risk (CVaR). However, VaR and CVaR (see, e.g., Rockafellar and Uryasev 2000) are typically used to measure potential losses, whereas the measures we define here measure worst-case revenues. Table 3 shows the relative percentage gap between Model 1 and the point estimate approach for each of these measures, based on one million test instances for each problem. A positive gap indicates that Model 1 has a higher value of the measure than the point estimate approach.

Based on these results, we conclude that Model 1 outperforms the point estimate approach on every variability and risk measure tested. The standard deviation of the revenue generated by Model 1 is in almost all cases less than or equal to that generated by the point estimate approach. The average revenue variability across all the test problems generated by Model 1 is much smaller than that generated by the point estimate approach. Similarly, except for very few cases, Model 1 has a higher worst case revenue, a higher maximum revenue target achievable at the 99% and 95% confidence levels, and a higher expected revenue over the worst 1% and 5% cases, than the point estimate approach. Consequently, at the same 99% or 95% confidence level, a higher revenue target can be set using Model 1 than using the point estimate approach, and the downside risk associated with Model 1 is less than that associated with the point estimate approach.

The above observations from the results in Tables 2 and 3 show that Model 1 is robust, and generates solutions that typically have low regret related measures, low variability and risk related measures, and high total expected revenue. As further validation for our results, we perform additional computational tests that use alternative ways to generate test instances. These include using a linear demand function instead of a constant elasticity model to generate base demand, and following a truncated normal distribution instead of a uniform distribution to generate actual demand realizations. Those additional tests generate very similar results to those reported above, and hence we do not report them in detail.

Our results support our intuition in proposing the minimax regret model. By minimizing the maximum regret, the model should consistently generate a pricing solution that is well balanced around the best possible solution under every possible demand realization. This guarantees good performance both with respect to expected revenue across all demand

Table 2 Performance of Models 1 and 2 Relative to Benchmark Approaches on Regret and Expected Revenue Measures

I_0	s_1, s_2	Maximum regret				Average relative regret					Total expected revenue			
		M2	PE	MM	ME	M1	M2	PE	MM	ME	M1	M2	PE	MM
S	0.05, 0.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	0.05, 0.1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	0.05, 0.2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	0.05, 0.4	0.00	8.16	57.89	8.16	0.56	2.23	0.36	4.30	0.36	0.65	3.27	0.00	5.48
	0.1, 0.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	0.1, 0.1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	0.1, 0.2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	0.1, 0.4	21.46	75.61	53.66	75.61	0.83	2.33	0.96	4.13	0.66	1.15	3.23	0.00	5.39
	0.2, 0.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	0.2, 0.1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	0.2, 0.2	0.01	0.00	138.09	0.00	0.01	0.01	0.03	0.58	0.04	0.00	0.00	0.00	0.38
	0.2, 0.4	15.69	64.23	63.56	64.23	2.65	5.53	4.11	4.37	2.11	2.11	8.05	0.00	6.38
	0.4, 0.05	85.97	34.84	93.21	34.84	0.64	0.30	0.96	0.64	0.30	0.07	0.00	0.00	0.07
	0.4, 0.1	0.00	0.00	0.00	28.57	0.98	3.40	1.76	1.82	0.64	0.10	3.14	0.04	0.31
	0.4, 0.2	0.00	24.94	53.79	41.95	3.15	5.52	2.87	4.64	3.10	5.54	8.05	0.11	6.02
	0.4, 0.4	17.56	64.65	51.64	69.28	4.79	5.42	6.80	6.85	7.71	6.22	9.84	0.25	12.94
	Mean	8.79	17.03	31.99	20.17	0.85	1.55	1.12	1.71	0.93	0.99	2.22	0.02	2.31
M	0.05, 0.05	0.00	20.36	0.00	20.36	0.75	1.75	1.30	1.75	1.30	0.46	0.46	0.00	0.46
	0.05, 0.1	8.97	29.12	15.38	29.12	1.26	5.42	2.70	3.86	2.70	1.69	2.90	0.50	1.27
	0.05, 0.2	5.42	42.88	5.42	27.85	3.53	8.25	6.10	6.39	5.34	3.99	5.83	1.14	3.82
	0.05, 0.4	6.57	68.68	32.29	68.68	3.54	5.63	10.57	7.77	10.57	3.16	5.36	0.51	9.60
	0.1, 0.05	15.55	58.49	15.55	37.98	0.95	5.42	3.30	2.95	2.49	0.57	3.19	0.91	0.57
	0.1, 0.1	0.00	30.39	0.00	21.39	2.26	6.88	4.54	5.26	3.42	2.02	3.78	1.17	2.02
	0.1, 0.2	4.67	45.51	4.67	35.65	3.82	9.24	6.62	7.13	5.81	3.42	6.46	2.40	4.04
	0.1, 0.4	9.82	44.92	29.11	67.78	3.37	6.44	11.32	8.11	11.42	4.81	5.39	0.35	9.26
	0.2, 0.05	4.84	23.42	4.60	6.14	1.02	8.79	4.97	4.49	3.47	0.58	7.67	2.26	2.01
	0.2, 0.1	6.60	22.10	5.46	24.16	2.69	9.20	4.91	5.72	4.43	1.42	7.31	1.17	1.44
	0.2, 0.2	0.30	21.34	4.94	12.76	3.94	8.47	7.49	8.27	6.68	2.21	6.50	1.49	4.55
	0.2, 0.4	8.10	45.98	27.37	42.30	4.90	8.22	13.20	10.25	12.31	6.65	5.74	3.59	11.70
	0.4, 0.05	0.00	32.44	6.29	6.66	1.08	5.22	5.23	4.34	4.31	1.31	8.23	1.03	1.37
	0.4, 0.1	0.00	0.56	12.94	9.61	2.16	6.03	5.03	5.91	5.41	1.68	7.95	1.15	5.61
	0.4, 0.2	6.34	34.27	22.56	12.52	5.54	7.77	8.73	8.38	7.99	2.00	7.35	1.36	7.02
	0.4, 0.4	6.44	37.75	28.14	18.61	11.21	12.67	14.50	11.30	15.65	2.71	6.65	2.49	10.78
	Mean	5.23	34.89	13.42	27.60	3.25	7.21	6.91	6.37	6.46	2.42	5.67	1.35	4.72
L	0.05, 0.05	0.00	5.26	0.00	5.26	2.27	3.27	1.25	3.27	1.23	2.07	2.07	0.82	2.07
	0.05, 0.1	8.81	27.54	8.81	27.54	2.29	5.61	2.77	4.31	2.76	1.69	3.09	0.60	1.73
	0.05, 0.2	10.81	45.80	10.81	35.29	4.38	9.55	6.29	7.38	5.83	4.18	6.47	0.39	4.18
	0.05, 0.4	16.76	51.42	16.09	51.42	5.47	12.87	12.98	12.13	13.34	3.27	10.36	1.01	9.56
	0.1, 0.05	13.43	19.98	15.43	24.78	1.88	5.63	2.42	2.85	2.40	0.50	3.51	1.02	0.52
	0.1, 0.1	0.00	8.74	0.00	17.99	2.73	6.80	3.57	4.73	3.52	1.52	3.84	1.07	1.52
	0.1, 0.2	5.09	36.11	5.09	31.24	4.70	10.01	6.88	7.87	6.38	3.78	6.51	1.29	4.04
	0.1, 0.4	8.11	45.10	16.55	45.10	7.96	13.36	13.66	12.84	14.30	8.41	10.41	1.11	9.93
	0.2, 0.05	4.72	4.84	0.00	4.84	2.97	9.50	4.07	3.97	4.06	0.67	7.71	1.01	0.67
	0.2, 0.1	5.25	36.09	2.14	18.51	3.77	9.95	6.01	5.86	5.09	1.57	7.35	1.80	1.65
	0.2, 0.2	1.40	34.93	5.54	13.72	4.72	9.84	8.59	8.03	7.73	2.44	6.61	2.03	4.84
	0.2, 0.4	4.86	36.60	9.05	28.41	8.13	13.08	12.12	11.68	15.12	3.39	10.56	2.19	9.42
	0.4, 0.05	1.92	13.11	0.00	13.11	2.61	10.98	6.65	4.61	6.65	0.78	11.08	1.01	0.78
	0.4, 0.1	0.00	30.23	5.05	14.28	2.72	11.42	10.79	4.40	7.52	1.31	10.78	1.78	4.31
	0.4, 0.2	2.52	36.16	6.62	13.40	4.38	11.14	14.49	6.18	9.30	2.22	10.12	2.30	5.92
	0.4, 0.4	2.26	20.98	5.07	20.98	7.77	11.20	14.10	7.68	15.29	3.89	11.26	1.22	8.69
	Mean	5.37	28.31	6.64	22.87	4.30	9.64	7.91	6.74	7.53	2.60	7.61	1.29	4.36

realizations and with respect to revenue under worst case demand realizations.

6. Concluding Remarks

This work considers dynamic pricing decisions that involve selling a given amount of inventory, without

replenishment, during a short selling season. We consider the selling season as composed of two periods, with demand subject to interval uncertainty. Our objective, which is minimizing the maximum total regret over the two periods, promotes robust decisions that are not extremely conservative. We consider a dynamic model as well as a static model of

Table 3 Performance of Model 1 and Point Estimate on Variability and Risk Measures

I_0	s_1, s_2	STD	WCR	99R	95R	99ER	95ER
S	0.05, 0.05	0.00	0.00	0.00	0.00	0.00	0.00
	0.05, 0.1	0.00	0.00	0.00	0.00	0.00	0.00
	0.05, 0.2	0.00	0.00	0.00	0.00	0.00	0.00
	0.05, 0.4	-6.44	1.13	1.13	1.01	1.13	0.70
	0.1, 0.05	0.00	0.00	0.00	0.00	0.00	0.00
	0.1, 0.1	0.00	0.00	0.00	0.00	0.00	0.00
	0.1, 0.2	0.00	0.00	0.00	0.00	0.00	0.00
	0.1, 0.4	-14.15	1.17	2.36	1.14	1.31	1.25
	0.2, 0.05	0.00	0.00	0.00	0.00	0.00	0.00
	0.2, 0.1	0.00	0.00	0.00	0.00	0.00	0.00
	0.2, 0.2	0.00	0.00	0.00	0.00	0.00	0.00
	0.2, 0.4	-10.59	1.39	1.37	2.71	1.38	2.24
	0.4, 0.05	10.10	1.29	1.29	0.05	1.29	0.59
	0.4, 0.1	4.20	-1.27	-1.27	-0.28	-1.27	-0.84
	0.4, 0.2	-38.78	1.22	1.18	1.07	1.38	1.27
	0.4, 0.4	-23.45	3.62	3.44	3.93	3.97	3.82
	Mean	-4.94	0.53	0.59	0.60	0.57	0.56
M	0.05, 0.05	-55.86	1.25	1.25	1.13	1.25	1.07
	0.05, 0.1	-13.63	2.52	2.45	1.19	2.52	1.97
	0.05, 0.2	-61.99	6.24	5.53	4.08	5.99	5.00
	0.05, 0.4	-7.73	5.00	4.31	4.12	4.89	4.73
	0.1, 0.05	0.50	3.30	2.76	1.70	3.23	2.42
	0.1, 0.1	-26.08	3.90	3.61	1.90	3.92	2.92
	0.1, 0.2	-53.55	4.80	4.73	5.36	4.61	4.69
	0.1, 0.4	-37.24	7.55	8.06	7.67	7.64	7.65
	0.2, 0.05	-9.73	3.18	2.62	1.42	3.01	2.11
	0.2, 0.1	-13.27	-0.47	0.40	1.04	-0.11	1.05
	0.2, 0.2	-19.41	-0.02	0.56	0.78	0.31	0.72
	0.2, 0.4	-35.44	6.17	6.87	8.18	6.45	7.36
	0.4, 0.05	2.85	1.71	1.75	1.83	1.82	1.86
	0.4, 0.1	-5.08	0.00	-0.05	-0.53	-0.06	-0.32
	0.4, 0.2	-3.09	6.94	6.63	3.97	7.01	5.52
	0.4, 0.4	14.70	-6.82	-7.16	-6.98	-6.99	-7.09
	Mean	-20.25	2.83	2.77	2.30	2.84	2.60
L	0.05, 0.05	-69.40	0.40	0.33	-0.13	0.35	0.19
	0.05, 0.1	-14.18	-0.83	-0.66	0.71	-0.80	0.31
	0.05, 0.2	-64.59	5.94	5.01	3.54	5.53	4.49
	0.05, 0.4	-26.72	2.64	2.53	2.68	2.68	2.67
	0.1, 0.05	-2.22	1.34	0.88	0.03	1.29	0.57
	0.1, 0.1	-31.40	1.20	1.15	1.54	1.22	1.38
	0.1, 0.2	-53.34	3.53	3.48	4.82	3.61	4.44
	0.1, 0.4	-66.88	4.01	6.11	8.82	5.02	7.66
	0.2, 0.05	3.34	1.79	0.96	-0.66	1.31	0.28
	0.2, 0.1	-4.04	1.16	1.53	2.61	1.27	2.02
	0.2, 0.2	-33.12	3.90	3.99	3.88	4.03	4.05
	0.2, 0.4	-30.36	5.35	5.10	4.53	5.26	4.98
	0.4, 0.05	12.80	1.68	2.02	1.98	1.90	2.10
	0.4, 0.1	18.42	4.80	5.73	5.98	5.42	5.87
	0.4, 0.2	-6.20	6.30	7.04	7.88	6.94	7.48
	0.4, 0.4	-9.55	3.97	4.20	5.04	4.12	4.57
	Mean	-23.59	2.95	3.09	3.33	3.07	3.32

this problem. For each model, we provide an efficient, optimal algorithm, based on solving systems of linear inequalities. Our computational results demonstrate that the prices generated by both our models are insensitive to errors in estimating the demand intervals. When compared to two other commonly used approaches that do not require demand probability

distributions, our dynamic model performs the best overall across a number of measures including maximum regret, average relative regret, variability, and risk measures. Further, our dynamic model generates a total expected revenue which closely approximates that of the maximum expected revenue approach which requires knowledge of demand probability distributions.

Several interesting issues remain open for future research. First, we conjecture that the minimax regret value is a nondecreasing function of the initial inventory, but this is not yet proved. Second, our interval uncertainty model could be modified to consider specific functional forms for demand, but with uncertainty about one or more parameters. Finally, our models could be generalized to consider multiple products with demand that is not independent, for example due to complementarity or substitutability. We hope that our work will encourage research on these important issues.

Appendix A. Solving Model 2

In this appendix, we describe the algorithm for solving problem P_1 defined in section 4.3 involving Model 2. To solve problem P_1 , we consider three cases:

Case 1: $I_2(p_2) \geq u_2(q_2)$ or $p_2 > q_2$.

In this case, we are in Case 1, 2.A or 2.B of Table 1, and hence $\Psi = \Psi_a$ as given in (22).

Case 1.1: $d_1(q_1) \leq I_0 - u_2(q_2)$ and $d_1(p_1) \leq I_0 - I_2(p_2)$.

We have

$$\Psi = \Psi_1 = q_1 d_1(q_1) - p_1 d_1(p_1) + [q_2 u_2(q_2) - p_2 I_2(p_2)]. \quad (A1)$$

Case 1.2: $d_1(q_1) > I_0 - u_2(q_2)$ and $d_1(p_1) \leq I_0 - I_2(p_2)$.

We have

$$\Psi = \Psi_2 = d_1(q_1)(q_1 - q_2) - p_1 d_1(p_1) + [q_2 I_0 - p_2 I_2(p_2)]. \quad (A2)$$

Case 1.3: $d_1(q_1) \leq I_0 - u_2(q_2)$ and $d_1(p_1) > I_0 - I_2(p_2)$.

We have

$$\Psi = \Psi_3 = q_1 d_1(q_1) + d_1(p_1)(p_2 - p_1) + [q_2 u_2(q_2) - p_2 I_0]. \quad (A3)$$

Case 1.4: $d_1(q_1) > I_0 - u_2(q_2)$ and $d_1(p_1) > I_0 - I_2(p_2)$.

We have

$$\Psi = \Psi_4 = d_1(q_1)(q_1 - q_2) + d_1(p_1)(p_2 - p_1) + [I_0(q_2 - p_2)]. \quad (A4)$$

Case 2: $I_2(p_2) < u_2(q_2)$, $p_2 \leq q_2$ and $p_1 \geq q_1$. This implies $d_1(p_1) \leq d_1(q_1)$ and (24) holds. Thus, Ψ_b can be written as

$$\Psi_b = q_1 d_1(q_1) - p_1 d_1(p_1) + (q_2 - p_2) \min\{I_0 - d_1(q_1), u_2(q_2)\}.$$

Case 2.1: $d_1(q_1) \leq I_0 - u_2(q_2)$.

In this case, $d_1(q_1) < I_0 - I_2(p_2)$. This, along with the fact that $I_2(p_2) < u_2(q_2)$, implies that $\min\{J_1, u_2(q_2)\} > I_2(p_2)$. Hence, we are in Case 2.C of Table 1, and $\Psi = \Psi_b$.

We have

$$\Psi = \Psi_5 = q_1 d_1(q_1) - p_1 d_1(p_1) + [(q_2 - p_2)u_2(q_2)]. \quad (A5)$$

Case 2.2: $I_0 - u_2(q_2) < d_1(q_1) < I_0 - I_2(p_2)$.

Again, it can be shown that we are in Case 2.C of Table 1. Hence, $\Psi = \Psi_b$ and we have

$$\Psi = \Psi_6 = d_1(q_1)(q_1 + p_2 - q_2) - p_1 d_1(p_1) + [(q_2 - p_2)I_0]. \quad (A6)$$

Case 2.3: $d_1(p_1) \leq I_0 - I_2(p_2) \leq d_1(q_1)$.

In this case, we are not in Case 2.C of Table 1, and hence $\Psi = \Psi_a$.

We have

$$\Psi = \Psi_2 = d_1(q_1)(q_1 - q_2) - p_1 d_1(p_1) + [q_2 I_0 - p_2 I_2(p_2)]. \quad (A7)$$

Case 2.4: $I_0 - I_2(p_2) < d_1(p_1) \leq d_1(q_1)$.

Again, we are not in Case 2.C of Table 1. Thus, $\Psi = \Psi_a$ and we have

$$\Psi = \Psi_4 = d_1(q_1)(q_1 - q_2) + d_1(p_1)(p_2 - p_1) + [I_0(q_2 - p_2)]. \quad (A8)$$

Case 3: $I_2(p_2) < u_2(q_2)$, $p_2 \leq q_2$ and $p_1 < q_1$. This implies $d_1(q_1) \leq d_1(p_1)$ and (25) holds. Note that for $p_1 < q_1$, Ψ_b can be written as

$$\Psi_b = q_1 d_1(q_1) - p_1 d_1(p_1) + q_2 \min\{I_0 - d_1(q_1), u_2(q_2)\} - p_2 \min\{I_0 - d_1(p_1), u_2(q_2)\}.$$

Case 3.1: $d_1(q_1) \leq I_0 - u_2(q_2)$
 $d_1(p_1) \leq I_0 - u_2(q_2)$.

and

In this case, $J_1 \geq u_2(q_2) > I_2(p_2)$. Hence, we are in Case 2.C of Table 1. Thus, $\Psi = \Psi_b$ and we have

$$\Psi = \Psi_7 = q_1 d_1(q_1) - p_1 d_1(p_1) + [u_2(q_2)(q_2 - p_2)]. \quad (A9)$$

Case 3.2: $d_1(q_1) \leq I_0 - u_2(q_2)$ and $d_1(p_1) > I_0 - u_2(q_2)$.

We are again in Case 2.C of Table 1. Thus, we have $\Psi = \Psi_b$ and

$$\Psi = \Psi_3 = q_1 d_1(q_1) + d_1(p_1)(p_2 - p_1) + [q_2 u_2(q_2) - p_2 I_0]. \quad (A10)$$

Case 3.3: $d_1(q_1) > I_0 - u_2(q_2)$.

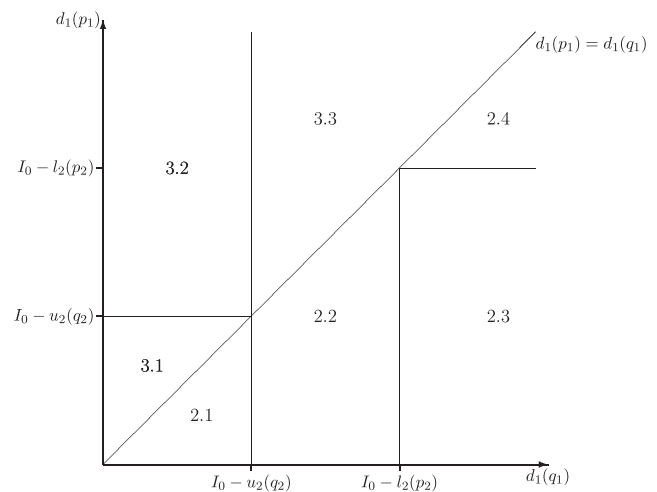
If $d_1(q_1) < I_0 - I_2(p_2)$, then we are in Case 2.C of Table 1. We have $\Psi = \Psi_b$. Also, $d_1(p_1) \geq d_1(q_1) > I_0 - u_2(q_2)$. Thus,

$$\Psi = \Psi_4 = d_1(q_1)(q_1 - q_2) + d_1(p_1)(p_2 - p_1) + [I_0(q_2 - p_2)]. \quad (A11)$$

If $d_1(q_1) > I_0 - I_2(p_2)$, then we are not in Case 2.C of Table 1, and hence $\Psi = \Psi_a$. From $p_1 < q_1$, we also have $d_1(p_1) \geq d_1(q_1) > I_0 - I_2(p_2) > I_0 - u_2(q_2)$. This implies $\Psi = \Psi_a = \Psi_4$, and we again have Equation (A11).

The above analysis shows that for given p_1, q_1, p_2, q_2 , we can partition the domain of $(d_1(q_1), d_1(p_1))$ into a constant number of subdomains x_i which represent the different cases for which Ψ is a linear function of $d_1(p_1)$ and $d_1(q_1)$. Under Case 1, the space of $(d_1(q_1), d_1(p_1))$ can be divided into four

Figure A1 Cases 2 and 3 of Model 2



subdomains which are quite straightforward. Under Cases 2 and 3, the space of $(d_1(q_1), d_1(p_1))$ can be divided as shown in Figure A1.

For any given values of p_1, p_2, q_1, q_2 , one of the three cases discussed above holds, and some or all of the associated subcases may hold. Problem P_1 can be solved by maximizing $\Psi(d_1(p_1), d_1(q_1))$ over the corresponding constraints of $(d_1(p_1), d_1(q_1))$ for each of the subcases that apply, and choosing the solution from the subcase that achieves the maximum Ψ . Given p_1, p_2, q_1, q_2 , for each of the subcases involved, the constraints of problem P_1 are defined by the inequalities (17)–(20) that apply, as well as the domain x_i of the relevant subcase, as discussed above.

Since each domain x_i is defined by linear inequalities, the solution of problem P_1 restricted to a specific case can be formulated as a linear program $LP(x_i)$ of the two variables $d_1(p_1)$ and $d_1(q_1)$. The face representing the constraints of $LP(x_i)$ consists of a rectangle that is intersected by at most one line $d_1(p_1) = d_1(q_1)$. Hence, such a face has at most five vertices and the optimal solution of $LP(x_i)$ can be found in constant time. Since the number of domains x_i is constant, the optimal value of $\tilde{Z}(p_1, q_1, p_2, q_2)$ can be found in constant time. Finally, we solve Model 2 using (21) by enumerating all possible values of p_1, p_2, q_1 and q_2 and performing a $\min_{p_1, p_2} \max_{q_1, q_2}$ optimization. There are $m_1^2 m_2^2$ different combinations for (p_1, p_2, q_1, q_2) .

Appendix B. Computational Results for Sensitivity Analysis

Table B1 below shows the computational results for the sensitivity analysis described in section 5.2.

Table B1 Sensitivity Analysis for Models 1 and 2

l_0	$(spread_1, spread_2)$	Model 1				Model 2			
		p_1^*	p_2^*	Max total regret	Period 1 regret	p_1^*	p_2^*	Max total regret	Period 1 regret
Small	0.05, 0.05	10	10	0	0	10	10	0	0
	0.05, 0.1	10	10	0	0	10	10	0	0
	0.05, 0.2	10	10	0	0	10	10	0	0
	0.05, 0.4	10	9	380	0	10	9	380	0
	0.1, 0.05	10	10	0	0	10	10	0	0
	0.1, 0.1	10	10	0	0	10	10	0	0
	0.1, 0.2	10	10	0	0	10	10	0	0
	0.1, 0.4	10	9	410	0	10	9	498	261
	0.2, 0.05	10	10	0	0	10	10	0	0
	0.2, 0.1	10	10	0	0	10	10	0	0
	0.2, 0.2	10	10	105	0	10	10	105	0
	0.2, 0.4	10	7	752	414	9	9	870	−208
	0.4, 0.05	10	9	221	0	10	10	411	411
	0.4, 0.1	10	9	560	0	10	9	560	0
	0.4, 0.2	9	9	870	−102	9	9	870	−102
	0.4, 0.4	9	8	1338	−102	9	8	1573	813
Medium	0.05, 0.05	5	7	786	−46	5	7	786	−46
	0.05, 0.1	6	5	1092	1244	6	5	1190	316
	0.05, 0.2	5	5	2029	−851	6	4	2139	8
	0.05, 0.4	6	4	2985	−489	5	5	3181	270
	0.1, 0.05	6	5	1048	166	6	5	1211	581
	0.1, 0.1	4	7	1599	−243	7	4	1599	1842
	0.1, 0.2	5	6	2314	658	5	5	2422	572
	0.1, 0.4	5	5	3250	−152	5	5	3569	658
	0.2, 0.05	5	5	2067	1347	5	5	2167	721
	0.2, 0.1	5	6	2272	1666	5	5	2422	1260
	0.2, 0.2	5	6	3322	1666	5	5	3332	1666
	0.2, 0.4	4	6	4234	1820	5	5	4577	1666
	0.4, 0.05	5	5	3181	2911	5	5	3181	2911
	0.4, 0.1	5	5	3569	2911	5	5	3569	2911
	0.4, 0.2	6	2	4304	3956	5	5	4577	2911
	0.4, 0.4	6	8	5470	3812	5	5	5822	2911
Large	0.05, 0.05	4	1	1008	1320	4	1	1008	1320
	0.05, 0.1	3	3	1271	865	3	2	1383	348
	0.05, 0.2	2	2	2284	−961	2	2	2531	−339
	0.05, 0.4	2	4	3742	405	1	2	4369	−1112
	0.1, 0.05	3	2	1251	756	3	2	1946	56

(continued)

Table B1 Continued

l_0	$(spread_1, spread_2)$	Model 1				Model 2			
		p_1^*	p_2^*	Max total regret	Period 1 regret	p_1^*	p_2^*	Max total regret	Period 1 regret
	0.1, 0.1	2	3	1946	56	2	3	1946	56
	0.1, 0.2	2	3	2711	814	2	2	2849	814
	0.1, 0.4	1	4	4266	930	1	2	4612	−1364
	0.2, 0.05	2	2	2417	1592	2	2	2531	780
	0.2, 0.1	2	3	2707	1951	2	2	2849	1485
	0.2, 0.2	2	3	3848	1951	2	2	3902	1951
	0.2, 0.4	2	3	5434	1951	1	2	5698	2110
	0.4, 0.05	2	1	4013	3588	2	1	4090	2880
	0.4, 0.1	2	1	4518	3588	2	1	4518	3588
	0.4, 0.2	2	1	5558	3448	2	1	5698	3588
	0.4, 0.4	2	1	7264	3448	1	2	7428	2923

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Appendix S1: Numerical Examples.